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Information and Communication Technologies 2
0796

GENERAL CERTIFICATE OF EDUCATION BOARD

General Certificate of Education Examination

JUNE 2024 **ADVANCED LEVEL**

Subject Title	**Information and Communication Technologies**
Subject Code No.	**0796**
Paper No.	**2**

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Duration: Two and a Half Hours

Answer any SIX questions.

All questions carry 17marks each. For your guidance, the approximate mark for each part of a question is indicated in brackets.

You will be graded on your ability to use good English, organise information clearly and the use of specialist vocabulary where appropriate.

In calculations, you are advised to show all the steps in your working, giving your answer at each stage.

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1. (i) (a) What is an information system? (1 mark)
(b) State and briefly explain the 5 basic components of an information system. (2 marks)
(c) Describe the Geographic Information System and explain one example of its application. (3 marks)

(ii) Define the following terms.
(a) Encryption. (2 marks)
(b) Prototype. (2 marks)
(c) Firewall. (2 marks)

(iii) (a) Describe the term Multimedia. (1 mark)
(b) Give one advantage and one disadvantage of social media. (2 marks)
(c) State two ethical behavior of proper use of social media. (2 marks)

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2. (i) The computer circuit is composed of logic gates, switches and wires.
(a) What is a logic gate? Give two examples of logic gates. (2 marks)
(b) Draw the symbols for the two gates mentioned above. (1 mark)
(c) State the role of a switch in a circuit. (1 mark)
(d) Represent the logical expression $\overline{A.B} + \bar{A}.\bar{B}$ by a circuit diagram. (3 marks)

(ii) Define the following terms as used in ICT.
(a) Computer ethics. (2 marks)
(b) Legislation. (1 mark)
(c) Ergonomics. (1 mark)

(iii) (a) Describe the term simulation and state one software that can be used to carry out simulation. (2 marks)
(b) Describe two applications or areas where simulation is useful. (2 marks)
(c) State one disadvantage and one advantage of simulation. (2 marks)

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3. (i) (a) State the functions of a graphic card in a computer. (2 marks)
(b) Explain the role of firmware in a system. (2 marks)
(c) Describe a device controller and give one example. (2 marks)

(ii) (a) Give two functions of the operating system. (2 marks)
(b) Explain pre-emptive and non-pre-emptive scheduling methods. (2 marks)
(c) State one property of each of command line and graphic user interfaces. Give an example of each of the interfaces. (3 marks)

(iii) (a) Describe a relational database. (2 marks)
(b) Differentiate between data redundancy and data integrity. (2 marks)

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4. (i) (a) Explain the term "network topology". (1 mark)
(b) Give two criteria that are considered in choosing a topology for a network. (2 marks)
(c) Explain why the ring topology has less fault tolerance than the bus topology. (2 marks)
(d) State THREE important resources of the network that are shared. (2 marks)

(ii) (a) Explain each of the following: spyware, virus, and worm. (3 marks)
(b) Explain two practices of protecting a computer from malware. (2 marks)
(c) State ONE manifestation that indicates the presence of a virus in a computer system. (1 mark)

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(iii) (a) Give the importance of file extension and state the name of the software that manages file extensions. **(2 marks)**

(b) Give the file extension for files created in:

- Microsoft excel.
- PowerPoint presentation.
- Notepad.
- Microsoft word. **(2 marks)**

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5. (i) (a) The effectiveness of any data communication system is based on THREE major components. State and explain the use of each of these components. **(3 marks)**

(b) Differentiate between synchronous and asynchronous data transmission. **(2 marks)**

(c) Describe the concept multiplexing as used in data communication. **(1 mark)**

(ii) (a) Define the term Internet protocol. **(1 mark)**

(b) State the importance of HTTP protocol. **(2 marks)**

(c) State the function of a router in a network. **(1 mark)**

(iii) (a) What is database normalization? **(2 marks)**

(b) State three advantages for normalizing a database. **(3 marks)**

(c) Give two properties of a table in second normal form. **(2 marks)**

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6. (i) (a) With reference to Algorithm, explain what you understand by the term peudocode. **(1 mark)**

(b) Describe a source code and an object code. **(2 marks)**

(c) Explain time complexity and space complexity with respect to algorithm design. **(2 marks)**

(ii) One type of system software is device driver.

(a) Describe device driver and give one example. **(2 marks)**

(b) State three other types of system software and give examples of each. **(3 marks)**

(iii) (a) What is computer ergonomics? **(1 mark)**

(b) Explain two measures that can be used to prevent RSI. **(2 marks)**

(c) Apart from RSI explain the cause and effects of two other health problems associated with the use of computers. **(2 marks)**

(iv) Define the following terms as use in ICT.

(a) Telemedicine. **(1 mark)**

(b) Videoconferencing. **(1 mark)**

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7. (i) One of the phases of SDLC is Feasibility Studies.

(a) List all the phases of SDLC in logical order. **(2 marks)**

(b) Explain two activities carried out by the system analyst at the feasibility phase. **(2 marks)**

(c) State and describe two conversion methods use to implement a new system in an organization. **(2 marks)**

(ii) Project managing activities are the sole responsibility of the project manager.

(a) List four activities carryout by the project manager. **(2 marks)**

(b) Describe two ways that project managers control projects. **(2 marks)**

(c) Explain why is it important to identify the critical path of a project? **(1 mark)**

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(iii) Carry out the following conversions.

(a) 56.12_{10} to binary. **(2 marks)**

(b) 10111101_2 to octal. **(2 marks)**

(c) $B73E_{16}$ to binary. **(2 marks)**

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8. (i) Define the following types of information systems.

(a) TPS **(2 marks)**

(b) MIS **(2 marks)**

(c) Expert system. **(2 marks)**

(ii) (a) Explain the term "digital divide". **(1 mark)**

(b) Is digital divide experienced among the students population? Explain your answer. **(2 marks)**

(c) Explain two ways digital divide can be reduced in Schools. **(2 marks)**

(iii) The **Web** is one of the services available on the **Internet**.

(a) Differentiate between the Internet and Web. **(2 marks)**

(b) Describe three other services available on the Internet. **(3 marks)**

(c) Describe E-Governance. **(1 mark)**

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